EPFL

1

Prof. Robert West
Machine Learning – CS-433 - MA
15.01.2026 from 15h15 to 18h15 in STCC
Duration: 180 minutes

Student 1

SCIPER : **999999**

Wait for the start of the exam before turning to the next page. This document is printed double sided, 20 pages. Do not unstaple.

- This is a closed book exam. No electronic devices of any kind.
- Place on your desk: your student ID, writing utensils, one double-sided A4 page cheat sheet if you have one; place all other personal items below your desk or on the side.
- You each have a different exam.
- This exam has many questions. We do *not* expect you to solve all of them even for the best grade
- Only answers in this booklet count. No extra loose answer sheets. You can use the last two pages as scrap paper.
- For the **multiple choice** questions, we give +2 points if your answer is correct, and 0 points for incorrect or no answer.
- For the **true/false** questions, we give +1.5 points if your answer is correct, and 0 points for incorrect or no answer.
- Use a **black or dark blue ballpen** and clearly erase with **correction fluid** if necessary.
- If a question turns out to be wrong or ambiguous, we may decide to nullify it.

Respectez les consignes suivantes | Observe this guidelines | Beachten Sie bitte die unten stehenden Richtlinien

choisir une réponse | select an answer Antwort auswählen	ne PAS choisir une réponse | NOT select an answer NICHT Antwort auswählen	Corriger une réponse | Correct an answer Antwort korrigieren
☒ ☑ ■	☐	☐

ce qu'il ne faut **PAS** faire | what should **NOT** be done | was man **NICHT** tun sollte

First part: multiple choice questions

For each question, mark the box corresponding to the correct answer. Each question has **exactly one correct answer**.

Question 1 Which of the following classification methods is non-parametric?

- ☐ Logistic Regression
- ☐ Linear decision boundaries
- ☐ Support Vector Machines
- ☐ Neural Networks
- ■ K-Nearest Neighbors

Solution: Every method assumes a particular functional form for the model, and hence, parametric. K-Nearest Neighbors do not assume a specific form for the prediction model, and hence, non-parametric.

Question 2 Consider the following one-dimensional logistic model on the data:

$$\mathbb{P}(y = 1|x; w) = \sigma(w \cdot x) = \frac{1}{1 + e^{-wx}}.$$

Assuming that the data is independent of the choice of w, what is the maximum likelihood estimate for w given observations $(x_1, y_1) = (1, 1)$ and $(x_2, y_2) = (-1, 1)$?

- ☐ $w = 0.5$
- ☐ $w = 1$
- ■ $w = 0$
- ☐ $w = -1$
- ☐ $w = -0.5$

Solution: By the assumption, we have $L(w) \propto \mathbb{P}(y = 1|x = 1; w)\mathbb{P}(y = -1|x = 1; w) = \sigma(w)\sigma(-w)$. Let $p = \sigma(w)$. Then, we have $1 - p = \sigma(-w)$. $L(w) = p(1 - p)$ is maximized when $p = \frac{1}{2}$ which implies that it is maximized for $w = 0$.

Question 3 Consider a binary classification problem where the observed feature $X \in \{0, 1, 2\}$ is generated from a latent variable $Z \in \{A, B\}$. The label is $Y \in \{0, 1\}$. The generative process is as follows:

(a) The latent variable Z is drawn according to

$$\mathbb{P}(Z = A) = 0.5, \qquad \mathbb{P}(Z = B) = 0.5.$$

(b) Conditional on Z, the label Y is drawn as

$$\mathbb{P}(Y = 1 \mid Z = A) = 0.75, \qquad \mathbb{P}(Y = 1 \mid Z = B) = 0.25.$$

(c) Conditional on (Y, Z), the observed feature X is drawn according to

$$\mathbb{P}(X = x \mid Y = y, Z = z) = \theta_{x,y,z},$$

where the parameters are given in the table below:

(y, z)	$X = 0$	$X = 1$	$X = 2$
$(0, A)$	0.5	0.25	0.25
$(1, A)$	0.25	0.5	0.25
$(0, B)$	0.5	0.25	0.25
$(1, B)$	0.25	0.25	0.5

Let $f_\star(x)$ be the Bayes classifier given the observation X and $g_\star(x, z)$ be the Bayes classifier given the observation X and the latent Z. Which of the following statement is **FALSE**?

- [ ] $\mathbb{P}(Y = 1 \mid X = 1) = \frac{7}{11}$
- [x] For any $x \in \{0, 1, 2\}$ and $z \in \{A, B\}$, $f_\star(x) = g_\star(x, z)$.
- [ ] $\mathbb{P}(Y = 0 \mid X = 0) = \frac{2}{3}$
- [ ] $f_\star(2) = 1$

Solution: We have the following equality:

$$\begin{aligned}
\mathbb{P}(Y = y \mid X = x) &= \frac{\mathbb{P}(Y = y, X = x)}{\mathbb{P}(X = x)} \\
&= \frac{\sum_{z \in \{A,B\}} \mathbb{P}(Y = y, X = x, Z = z)}{\sum_{z \in \{A,B\}} \sum_{y \in \{0,1\}} \mathbb{P}(Y = y, X = x, Z = z)} \\
&= \frac{\sum_{z \in \{A,B\}} \mathbb{P}(Z = z)\mathbb{P}(Y = y \mid Z = z)\mathbb{P}(X = x \mid Y = y, Z = z)}{\sum_{z \in \{A,B\}} \sum_{y \in \{0,1\}} \mathbb{P}(Z = z)\mathbb{P}(Y = y \mid Z = z)\mathbb{P}(X = x \mid Y = y, Z = z)}.
\end{aligned}$$

Since, $\mathbb{P}(Z = A) = \mathbb{P}(Z = b)$, we derive:

$$\mathbb{P}(Y = y \mid X = x) = \frac{\sum_{z \in \{A,B\}} \mathbb{P}(Y = y \mid Z = z)\mathbb{P}(X = x \mid Y = y, Z = z)}{\sum_{z \in \{A,B\}} \sum_{y \in \{0,1\}} \mathbb{P}(Y = y \mid Z = z)\mathbb{P}(X = x \mid Y = y, Z = z)}.$$

We derive the following expressions:

$$\begin{aligned}
\mathbb{P}(Y = 0 \mid X = 0) &= \frac{0.25 \times 0.5 + 0.75 \times 0.5}{0.25 \times 0.5 + 0.75 \times 0.25 + 0.75 \times 0.5 + 0.25 \times 0.25} = \frac{2}{3}, \\
\mathbb{P}(Y = 1 \mid X = 1) &= \frac{0.75 \times 0.5 + 0.25 \times 0.25}{0.25 \times 0.25 + 0.75 \times 0.5 + 0.75 \times 0.25 + 0.25 \times 0.25} = \frac{7}{11} \\
\mathbb{P}(Y = 1 \mid X = 2) &= \frac{0.75 \times 0.25 + 0.25 \times 0.5}{0.25 \times 0.25 + 0.75 \times 0.25 + 0.75 \times 0.25 + 0.25 \times 0.5} = \frac{5}{9}.
\end{aligned}$$

For the wrong choice, we do the same computation:

$$\begin{aligned}
\mathbb{P}(Y = y \mid X = x, Z = z) &= \frac{\mathbb{P}(Y = y, X = x, Z = z)}{\mathbb{P}(X = x, Z = z)} \\
&= \frac{\mathbb{P}(Y = y, X = x, Z = z)}{\sum_{y \in \{0,1\}} \mathbb{P}(Y = y, X = x, Z = z)}
\end{aligned}$$

Question 4 (Loss Function) Suppose that the teaching team of the ML course wants to develop an autograder that takes into account the scores for each project, and frequency of class attendance and predict student performance on this final (between 0 and 100). Which of the following functions is an appropriate loss function for this problem?

- [x] Mean Absolute Error Loss
- [ ] Logistic Loss
- [ ] Hinge Loss
- [ ] 0-1 Loss

Solution: The problem is a regression problem since the output is continuous (0-100). Therefore, Mean Absolute Error Loss is an appropriate loss function.

Question 5 (Gradient of Linear MSE) Consider a linear regression model $y = Xw$, with the mean squared error loss function:

$$L(w) = \frac{1}{2N} \sum_{n=1}^{N} (y_n - x_n^\top w)^2$$

, Which of the following expressions correctly represents the gradient of $L(w)$ with respect to w?

- [ ] $\frac{1}{N} X(y - Xw)$
- [ ] $\frac{1}{N} X^\top (y - Xw)$
- [x] $-\frac{1}{N} X^\top (y - Xw)$
- [ ] $X^\top Xw - X^\top y$

Solution: Correct answer: (b). A and C are clearly wrong. Between B and D, the sign is the difference.

Question 6 (Computational Complexity of the Hessian) Consider a linear regression problem with N training samples and D parameters, and the mean squared error loss

$$L(w) = \frac{1}{2N} \sum_{n=1}^{N} (y_n - x_n^\top w)^2.$$

The Hessian matrix $\nabla^2 L(w)$ contains all second derivatives of $L(w)$ with respect to the parameters. What is the computational complexity (in big-O notation) of computing the Hessian matrix using standard matrix multiplication?

- [ ] $O(ND)$
- [x] $O(ND^2)$
- [ ] $O(N^2D)$
- [ ] $O(D^2)$

Solution: For this loss the Hessian is

$$\nabla^2 L(w) = \frac{1}{N} \sum_{n=1}^{N} x_n x_n^\top = \frac{1}{N} X^\top X$$

, which is a $D \times D$ matrix. Computing each entry of $X^\top X$ requires summing over N samples, and there are $O(D^2)$ entries, so the total cost is $O(ND^2)$.

Question 7 (Ridge Regression and High Dimensionality) Consider the linear regression model $y = Xw + \varepsilon$ with $X \in \mathbb{R}^{n\times d}$. As the ratio d/n approaches 1, and especially when $d > n$, which of the following statements are correct? (Select all that apply.)

- ☐ Ridge regression suffers from the same non-invertibility issue as OLS and provides no improvement.
- ■ Adding ridge regularization $\lambda\|w\|_2^2$ ensures that $X^\top X + \lambda I$ is invertible, even when $d > n$.
- ☐ Ordinary least squares (OLS) can still compute a unique solution because $X^\top X$ remains invertible.
- ☐ When $d > n$, the matrix $X^\top X$ becomes singular and the OLS admits no solutions

Solution: Correct answers: (d). For $d > n$, $X^\top X$ is singular and OLS has many solution, while ridge regression stabilizes the problem by adding λI, making the system invertible.

Question 8 (Least Squares Orthogonal Design Matrix) Consider the least-squares estimator

$$w^\star = (X^\top X)^{-1} X^\top y,$$

where $X \in \mathbb{R}^{n\times d}$ and $y \in \mathbb{R}^n$. Assume that $n = d$ and that all feature vectors are orthogonal, i.e.

$$X^\top X = cI_d$$

for some constant $c > 0$. Let x_m denote a test point.
Which of the following statements are correct? (Select all that apply.)

- ■ Each weight w_j depends only on the corresponding training output along its coordinate direction.
- ■ The solution simplifies to $w^\star = \frac{1}{c} X^\top y$, which can be seen as an orthogonal decomposition.
- ■ For any test point x_m, the prediction is $\hat{y}_m = \frac{1}{c} x_m^\top X^\top y$.
- ■ The Gram matrix $X^\top X$ is invertible, and its inverse is $(1/c)I_d$.
- ☐ Each weight w_j depends on all training outputs y_i.

Solution: Correct answers: (a), (b), (d), (e). Since $X^\top X = cI_d$, the inverse is $(1/c)I_d$, giving

$$w^\star = \frac{1}{c} X^\top y.$$

Each coordinate of $w^\star$ depends only on its aligned feature vector (no cross-coupling). The prediction for a test input is

$$\hat{y}_m = x_m^\top w^\star = \frac{1}{c} x_m^\top X^\top y.$$

Question 9 (SVM) Which of the following is NOT an advantage of SVMs?

- ☐ Effective in high-dimensional spaces
- ■ Naturally outputs probabilities for each class
- ☐ Robust to overfitting (with proper kernel and regularization)
- ☐ Memory efficient when number of support vectors is small

Solution: SVM does not output probabilities for each class naturally.

Question 10 (L1-SVM) Standard soft SVM has the following optimization problem (C is a constant):

$$\min_w \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n} \max(0, 1 - y_i(X_i^\top w))$$

Now if we replace the L2 regularization term with L1 regularization, i.e.,

$$\min_w \frac{1}{2}\|w\|_1 + C\sum_{i=1}^{n} \max(0, 1 - y_i(X_i^\top w))$$

Which of the following statements is NOT true for L1-SVMs?

- [ ] L1-SVM is no longer a smooth optimization problem.
- [x] The dual problem of the L1-SVM can be easily expressed in terms of kernel functions.
- [ ] L1-SVM remains a convex problem, which is still globally solvable.
- [ ] L1-SVM can perform feature selection by driving some weights to zero.

Solution: L1-SVM is convex and has sparse solutions. L1-SVM is not a smooth optimization problem due to the presence of the $L1$ norm, which is not differentiable at zero. The dual problem of L1-SVM is not quadratic and depends directly on x_{ij}s, and thus is not directly kernel compatible. Note the dual derivation is tricky, but the other 3 options are quite easy to verify.

Question 11 (Receptive Field) Consider a 1D sequence $X \in \mathbb{R}^{T\times 1}$ of length T. Assume T is even, and $T > 100$. What is the receptive field of the output at position $Y[T/2]$?
Assume the 1D sequence is processed as follows: $Y = f \circ g \circ h(X)$ (first apply h, then g, then f).

- h is a 1D convolution with kernel size 3, padding 1, and stride 1
- g is a self-attention layer $\mathrm{attn}(X, X, X, 1)$
- f is a 1D convolution with kernel size 5, padding 2, and stride 2

As a reminder, the self-attention operation is given by

$$\mathrm{attn}(Q, K, V, d) = \text{row-wise-softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V.$$

- [ ] $T/2$
- [x] T
- [ ] 5
- [ ] 7
- [ ] 9
- [ ] 3

Solution: The receptive field is T at every position, since all positions interact with each other during self-attention.

Question 12 (Regularization) Regularization usually refers to techniques used to reduce overfitting. Which of the following should *not* be considered a regularization method?

- [ ] Horizontal flip (images)
- [x] Adam
- [ ] Weight decay
- [ ] Dropout

Solution: Adam is an optimization algorithm, not a regularization method.

Question 13 (Forward Pass Complexity) Consider the following layer operation: $Y = [\sigma(XW_1) \odot \sigma(XW_2)]W_3$, where:

- $X \in \mathbb{R}^{n \times K}$
- $W_1, W_2 \in \mathbb{R}^{K \times H_1}$
- $W_3 \in \mathbb{R}^{H_1 \times H_2}$
- $\sigma(\cdot)$ is an element-wise activation function
- $\odot$ denotes point-wise multiplication (think `*` in numpy)

How many scalar multiplications are needed to implement the forward pass of this layer?

- ☐ $2nKH_1 + nH_1 + H_1H_2$
- ■ $2nKH_1 + nH_1 + nH_1H_2$
- ☐ $2nKH_1 + nH_1H_2$
- ☐ $nKH_1 + nH_1 + nH_1H_2$
- ☐ $2nKH_1 + nK + nH_1H_2$

Solution:

Question 14 (Attention) Recall that the output of self-attention is given by the following formula:

$$Z = f(X) = \text{softmax}\left(\frac{XW_QW_K^\top X^\top}{\sqrt{d_k}}\right)XW_V\,,$$

where $X \in \mathbb{R}^{n\times d}, W_Q \in \mathbb{R}^{d\times d_k}, W_K \in \mathbb{R}^{d\times d_k}, W_V \in \mathbb{R}^{d\times d}$ are the data, query, key, and value matrices, respectively. Here, d is the dimensionality of the tokens, d_k is the hidden dimensionality, n is the sequence length. The softmax is applied row-wise. Additionally, the dimensions satisfy $d_k < d < n$.
Let $R \in \{0,1\}^{n\times n}$ be a permutation matrix, i.e, it satisfies $R^\top R = RR^\top = I_n$. Consider the reordered input tokens $X' = RX$. Which of the following mathematical statements correctly describes the relationship between the new output $Z' = f(X')$ and the original output Z?

- [ ] $Z' \neq RZ$ and $Z' \neq Z$
- [x] $Z' = RZ$
- [ ] $Z' = ZR^\top$
- [ ] $Z' = Z$

Solution: Let $Q = XW_Q$, $K = XW_K$ and $V = XW_V$.
We then have

$$\begin{aligned} Z' = f(X') &= \text{softmax}\left(\frac{X'W_QW_K^\top X'^\top}{\sqrt{d_k}}\right)X'W_V \\ &= \text{softmax}\left(\frac{RXW_QW_K^\top X^\top R^\top}{\sqrt{d_k}}\right)RXW_V \\ &= \text{softmax}\left(\frac{RQK^\top R^\top}{\sqrt{d_k}}\right)RV \end{aligned}$$

Since softmax is applied row wise, the permutation has no effect inside the operation:

$$\text{softmax}\left(\frac{RQK^\top R^\top}{\sqrt{d_k}}\right)RV = R\text{softmax}\left(\frac{QK^\top R^\top}{\sqrt{d_k}}\right)RV = R\text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)R^\top RV$$

Since $R^TR = I$:

$$Z' = R\text{softmax}\left(\frac{QK^T}{\sqrt{D_K}}\right)V = RZ$$

Therefore, answer is $Z' = RZ$.

Question 15 In a decoder-only model, the model must be prevented from "cheating" by seeing future words. This is enforced mathematically by adding a causal mask matrix $M \in \mathbb{R}^{n\times n}$ to the self-attention operation of above:

$$Z = f(X) = \text{softmax}\left(\frac{XW_QW_K^\top X^\top + M}{\sqrt{d_k}}\right)XW_V\,,$$

Which is the correct definition of the matrix M?

- [ ] $M_{ij} = M_{ji}$ for all i, j.
- [ ] $M_{ij} = -\infty$ if $j < i$ and $M_{ij} = 0$ otherwise.
- [x] $M_{ij} = -\infty$ if $j > i$ and $M_{ij} = 0$ otherwise.
- [ ] $M_{ij} = -\infty$ if $i \neq j$ and $M_{ij} = 0$ otherwise.

Solution: The causal mask ensures that token i can only attend to tokens at positions $j \leq i$ (i.e., itself and previous tokens). This is achieved by setting $M_{ij} = -\infty$ for $j > i$, which causes the softmax to output zero attention weight for future positions. When $j \leq i$, we set $M_{ij} = 0$ to leave the attention scores unchanged. This makes M a lower triangular matrix (with zeros on and below the diagonal, and $-\infty$ above).
This ensures autoregressive generation where each token can only depend on past context.

Question 16 Which of the following statements is **TRUE** about the influence of the causal mask matrix M on the complexity of the self-attention operation, expressed in terms of the sequence length n?

- ☐ It keeps the complexity of the self-attention operation at $\mathcal{O}(n)$.
- ☐ It reduces the complexity of the self-attention operation from $\mathcal{O}(n^2)$ to $\mathcal{O}(n)$.
- ■ It keeps the complexity of the self-attention operation at $\mathcal{O}(n^2)$.
- ☐ It increases the complexity of the self-attention operation from $\mathcal{O}(n)$ to $\mathcal{O}(n^2)$.

Solution: The causal mask matrix has no effect on the complexity of the self-attention operation, it is still $\mathcal{O}(n^2)$.

Question 17 (Adversarial Risk) Consider the definition of adversarial risk:

$$R_\varepsilon(f) = \mathbb{E}_{(x,y)\sim\mathcal{D}}\left[\max_{\hat{x}, \|\hat{x}-x\|\leq\varepsilon} \mathbb{1}_{f(\hat{x})\neq y}\right],$$

which changes from the standard risk $R(f) = \mathbb{E}_{(x,y)\sim\mathcal{D}}\left[\mathbb{1}_{f(x)\neq y}\right]$ by considering the worst-case perturbation within an ε-ball.
Which of the following statements is **TRUE**?

- ☐ If $R(f) = 0$ (zero standard risk), then $R_\varepsilon(f) = 0$ necessarily.
- ■ $R_\varepsilon(f) \geq R(f)$ always holds for any $\varepsilon > 0$.
- ☐ Adversarial risk is independent of the choice of norm used to measure $\|\hat{x} - X\|$.
- ☐ The adversarial risk is independent of the data distribution $\mathcal{D}$.

Solution: Standard risk $R(f)$ evaluates misclassification on clean data only. Adversarial risk $R_\varepsilon(f)$ evaluates misclassification under worst-case perturbations within an ε-ball. Since the clean input X is contained in this ball (the perturbation can be zero), the adversary's optimization problem includes the original input as a feasible point. Therefore, the maximum over the perturbed set is at least as large as the loss on the clean input, yielding $R_\varepsilon(f) \geq R(f)$.
For the other options: perfect accuracy on clean data does not guarantee robustness. Different norms define different perturbation geometries and yield different adversarial risks. The adversarial risk is dependent on the data distribution $\mathcal{D}$ which defines the expectation.

Question 18 (Fairness) A bank is training a machine learning model to predict creditworthiness ($Y \in \{0,1\}$). The dataset contains a sensitive attribute A (e.g., race) and several other features X (e.g., income, zip code, profession). The engineers decide to remove the column A from the training data to ensure the model cannot discriminate based on race.
Which of the following statements best describes the effectiveness of this strategy?

- ☐ This strategy is effective because the model cannot learn a correlation with A if A is not in the input.
- ☐ This strategy is ineffective because it causes the model to overfit to the minority class.
- ■ This strategy is ineffective because the remaining features X are likely to be correlated with A in a way that allow the model to make a decision based on A.
- ☐ This strategy is effective, provided that the dataset is balanced (equal number of samples from all groups).

Solution: Removing sensitive attributes does not solve the fairness problem because other features can serve as proxies for the sensitive attributes, allowing the model to reconstruct the sensitive information.

Question 19 (K-means) Given a dataset $\mathcal{D} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ with $\boldsymbol{x}_i \in \mathbb{R}^d$, the K-means problem aims to find the optimal cluster centers $C^* = \{\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_K\}$ given by

$$C^* = \arg \min_{C=\{\boldsymbol{\mu}_1,\ldots,\boldsymbol{\mu}_K\}} \mathcal{R}(C) \quad \text{where } \mathcal{R}(C) = \sum_{\boldsymbol{x}_i \in \mathcal{D}} \min_{z_i} \|\boldsymbol{x}_i - \boldsymbol{\mu}_{z_i}\|_2^2,$$

where $z_i \in \{1, \ldots, K\}$ with $K \geq 2$ indicates the cluster assigned to data point $\boldsymbol{x}_i$. In this problem, we consider the K-means algorithm (also known as Lloyd's algorithm) you covered in class as an approach for solving the K-means problem. Assume that the K-means algorithm terminates once $\mathcal{R}(C)$ did not improve for one iteration.
To determine a good choice of K, we run K-means for $K \in \{2, \ldots, n\}$. What trend do we expect to find as we increase K?

- ☐ $\mathcal{R}(C^*)$ initially increases and then decreases.
- ■ $\mathcal{R}(C^*)$ decreases.
- ☐ $\mathcal{R}(C^*)$ initially decreases and then increases.
- ☐ $\mathcal{R}(C^*)$ increases.

Solution: The objective function $\mathcal{R}(C)$ represents the sum of squared Euclidean distances between each data point $\boldsymbol{x}_i$ and its assigned cluster center $\boldsymbol{\mu}_{z_i}$. As we increase the number of clusters K:

- The algorithm has more centers to place within the data space.
- Any data point $\boldsymbol{x}_i$ will either remain assigned to its previous center or be reassigned to a new center if the new center is closer.
- Consequently, the distance $\|\boldsymbol{x}_i - \boldsymbol{\mu}_{z_i}\|_2^2$ for any given point can only remain the same or decrease.

Therefore, the total error $\mathcal{R}(C^*)$ decreases monotonically as K increases. In the extreme case where $K = n$ (where n is the number of data points), every data point would be its own cluster center, resulting in $\mathcal{R}(C^*) = 0$.

Question 20 (EM) Which of the following statements about the Expectation-Maximization (EM) algorithm is **FALSE**?

- ☐ An EM algorithm can also be used to fit GMMs in the semi-supervised setting, where some data points are labeled and some are unlabeled.
- ☐ The log-likelihood of the observed data increases (or stays constant) monotonically with each iteration of the soft EM algorithm.
- ☐ The objective function (log-likelihood) for Gaussian Mixture Models is generally non-convex, meaning EM may converge to a local optimum.
- ■ For fitting GMMs to data with a lot of overlapping clusters, hard EM typically performs better than soft EM.

Solution: EM works well for semi-supervised learning by fixing the cluster assignments for labeled data and estimating them for unlabeled data. A key property of EM is that the likelihood function never decreases after an iteration (monotonicity). Hard EM (similar to K-Means) makes binary assignments. When clusters overlap significantly, there is high uncertainty. **Soft EM** performs better in these cases because it assigns probabilistic cluster assignments (e.g., 60% cluster A, 40% cluster B), allowing the model to capture the ambiguity and overlap more accurately. The GMM objective is non-convex, so initialization matters and global optimality is not guaranteed.

Question 21 (Matrix Factorization) Consider a recommender system problem where we aim to approximate a rating matrix $X \in \mathbb{R}^{D\times N}$ (where D is the number of items and N is the number of users) using two low-rank factor matrices $W \in \mathbb{R}^{D\times K}$ and $Z \in \mathbb{R}^{N\times K}$. The goal is to minimize the following regularized objective function over the set of observed entries Ω:

$$\min_{W,Z} \mathcal{L}(W, Z) := \frac{1}{2} \sum_{(d,n)\in\Omega} [x_{dn} - (WZ^\top)_{dn}]^2 + \frac{\lambda_w}{2}\|W\|_F^2 + \frac{\lambda_z}{2}\|Z\|_F^2$$

Which of the following statements accurately describes the **Alternating Least-Squares (ALS)** algorithm for solving this problem?

- ☐ The ALS algorithm updates both matrices W and Z simultaneously using the gradients $\nabla_W\mathcal{L}$ and $\nabla_Z\mathcal{L}$ to find the global minimum of the jointly convex objective.
- ■ The ALS algorithm exploits the fact that if we fix one matrix (e.g., Z), the optimization problem with respect to the other matrix (e.g., W) becomes a convex quadratic (Least Squares) problem with a closed-form solution.
- ☐ The ALS algorithm is strictly limited to cases where the matrix X is fully observed (i.e., there are no missing ratings), as the closed-form updates cannot be derived for sparse matrices.
- ☐ The ALS algorithm is a specific type of Stochastic Gradient Descent (SGD) that updates the parameters based on the error of a single randomly selected rating (d, n) at each step.

Solution: The objective function is *not* jointly convex with respect to W and Z due to the product term $WZ^\top$. s described in the lecture, ALS works by "coordinate descent": it minimizes the cost w.r.t. Z for fixed W and then minimizes w.r.t. W given Z. When one matrix is fixed, the problem becomes a linear least squares problem (Ridge Regression), which is convex and has a closed-form solution. This describes Stochastic Gradient Descent (SGD), which is a different method discussed separately in the lectures. While the basic derivation often assumes a full matrix for simplicity, ALS can be (and commonly is) derived for the general setting with missing entries, as noted in the slides discussing "ALS with Missing Entries".

Question 22 (3D Parallelism) You are training an LLM on EPFL's RCP cluster with $N_{\text{total}} = 512$ GPUs. Your configuration uses $N_{\text{TP}} = 8$ (Tensor Parallelism) and $N_{\text{PP}} = 4$ (Pipeline Parallelism). If you accumulate $A = 32$ micro-batches and the per-device micro-batch size is $B_{\text{micro}} = 2$. what is the Global Batch Size (B_{global}) for a single optimization step?

- ☐ $B_{\text{global}} = 128$
- ☐ $B_{\text{global}} = 2048$
- ☐ $B_{\text{global}} = 32768$
- ■ $B_{\text{global}} = 1024$
- ☐ $B_{\text{global}} = 512$

Solution: $B_{\text{global}} = N_{\text{DP}} \times A \times B_{\text{micro}} = 16 \times 32 \times 2 = 1024$
where $N_{\text{DP}} = \frac{N_{\text{total}}}{N_{\text{TP}}\times N_{\text{PP}}} = \frac{512}{8\times 4} = 16$
Note: N_{TP} and N_{PP} split the model across GPUs, so they don't increase the global batch size. They only determine how many data-parallel replicas (N_{DP}) can fit in the 512 GPUs.

Question 23 (FastText) Consider a simplified FastText model with embedding dimension $K = 1$. The score is given by $s = w(\mathbf{z}^\top \mathbf{x}_n)$, where $\mathbf{x}_n$ is the bag-of-words vector. Let $\mathbf{g}_A$ and $\mathbf{g}_B$ be the gradients of the loss w.r.t. $\mathbf{z}$ for two sentences. Assuming $w \neq 0$ and non-zero error, $\langle \mathbf{g}_A, \mathbf{g}_B \rangle = 0$ if and only if:

- [x] The sentences share no common words.
- [ ] The parameter vector $\mathbf{z}$ is zero.
- [ ] The sentences have the same length.
- [ ] The sentences are identical.

Solution: The gradient of the loss w.r.t. $\mathbf{z}$ is proportional to $w \cdot \mathbf{x}_n$ (by chain rule).
For two sentences A and B with bag-of-words vectors $\mathbf{x}_A$ and $\mathbf{x}_B$:

$$\langle \mathbf{g}_A, \mathbf{g}_B \rangle \propto w^2 \cdot \langle \mathbf{x}_A, \mathbf{x}_B \rangle$$

Since $w \neq 0$, this inner product is zero if and only if $\langle \mathbf{x}_A, \mathbf{x}_B \rangle = 0$.
For bag-of-words vectors, $\langle \mathbf{x}_A, \mathbf{x}_B \rangle = 0$ means the sentences share no common words (each shared word would contribute a positive term to the inner product).

Question 24 (Cross-Validation) A student is trying to perform model selection for a ridge regression model. They want to choose the best hyperparameter λ from a set of $N = 5$ different candidate values $\{\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5\}$. To evaluate these candidates, they decide to use the K-fold cross-validation procedure, setting $K = 5$. To find the best λ, what is the **total number of times** the learning algorithm must be trained?

- [x] 25
- [ ] 1
- [ ] 5
- [ ] 10

Solution: The cross-validation should be performed on each of the hyperparameters choices $\{\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5\}$ individually. For each 5-fold cross validation we should train the learning algorithm 5 times, each time holding out a $\frac{1}{5}$-th of data. Thus, the correct answer is 25.

Question 25 (Bias-Variance Decomposition) Which of the following statements **correctly** describes the bias, variance, and noise components in the bias-variance decomposition?

- [ ] A highly complex model (e.g., a high-degree polynomial) has high **Bias** because it "overfits" the data, and low **Variance** because it fits the training points well.
- [ ] **Bias** measures a variability of a model's prediction when trained on different datasets from the same distribution.
- [x] The **Noise** term, represents the irreducible error, and is the only component that does not depend on the choice of the learning algorithm or model complexity.
- [ ] The **Variance** term, measures the difference between the true function $f(x)$ and the average prediction $\mathbb{E}_S[f_S(x)]$ over all possible training sets.

Solution: The definition given for variance, is for bias, and vice versa. Also, a highly complex model has a high variance and low bias. The statement about the noise term is the only correct one.

Question 26 (Kernel Methods)

Which of the following functions is a **valid positive-semidefinite kernel** on $\mathbb{R}^n$?

- ☐ $k(x, y) = -\|x - y\|^2$
- ☐ $k(x, y) = |x^\top y|$
- ☐ $k(x, y) = -x^\top y$
- ☐ $k(x, y) = 1 - x^\top y$
- ■ $k(x, y) = (1 + x^\top y)^2$

Solution: **Correct answer:** $(1 + x^\top y)^2$.

Why $(1 + x^\top y)^2$ is PSD. Expand:

$$(1 + x^\top y)^2 = 1 + 2(x^\top y) + (x^\top y)^2.$$

Each term is a PSD kernel:
- 1 is the constant kernel, - $x^\top y$ is the linear kernel, - $(x^\top y)^2$ is the degree-2 homogeneous polynomial kernel. A nonnegative sum of PSD kernels is PSD. Hence $(1 + x^\top y)^2$ is a valid kernel.

Why $|x^\top y|$ is *not* PSD. Consider two points $x_1 = v$ and $x_2 = -v$ with $v \neq 0$. Compute the Gram matrix:

$$K = \begin{pmatrix} |v^\top v| & |v^\top(-v)| \\ |(-v)^\top v| & |(-v)^\top(-v)| \end{pmatrix} = \begin{pmatrix} \|v\|^2 & \|v\|^2 \\ \|v\|^2 & \|v\|^2 \end{pmatrix}.$$

This matrix has eigenvalues

$$2\|v\|^2 \quad \text{and} \quad 0,$$

so far it is PSD; however, introduce a third point $x_3 = 2v$. Then the 3×3 Gram matrix becomes rank-1 with structure:

$$K_{ij} = |x_i^\top x_j|.$$

A direct calculation shows that its determinant is negative for generic v because the inner products do not preserve the sign structure needed for positive semidefiniteness. (Explicitly: $\det K = -2\|v\|^6 < 0$.) Thus $|x^\top y|$ fails the PSD condition.

Why $1 - x^\top y$ is not PSD. Take $x_1 = v$, $x_2 = -v$. Then

$$K = \begin{pmatrix} 1 - \|v\|^2 & 1 + \|v\|^2 \\ 1 + \|v\|^2 & 1 - \|v\|^2 \end{pmatrix}.$$

Its eigenvalues are

$$(1 - \|v\|^2) \pm (1 + \|v\|^2),$$

giving a negative eigenvalue when $\|v\| > 0$. Therefore it is not PSD.

Why $-\|x - y\|^2$ is not PSD. For $x_1 = 0$ and $x_2 = v \neq 0$:

$$K = \begin{pmatrix} 0 & -\|v\|^2 \\ -\|v\|^2 & 0 \end{pmatrix},$$

with eigenvalues $\pm\|v\|^2$, so it is indefinite.

Why $-x^\top y$ is not PSD. Even for a single nonzero vector v,

$$K = \begin{pmatrix} -\|v\|^2 & -\|v\|^2 \\ -\|v\|^2 & -\|v\|^2 \end{pmatrix}$$

has a negative eigenvalue $-\|v\|^2$. Thus it is not PSD.

Question 27 (ReLU Networks) You are designing a neural network with a single hidden layer using ReLU activations, denoted as $(z)_+ = \max\{0, z\}$. The network must exactly represent a piecewise linear function $f(x)$ defined by the following slopes:

- For $x < 2$: the slope is 0 (and $f(x) = 0$).
- For $2 \leq x < 5$: the slope is 3.
- For $x \geq 5$: the slope is -1.

Which of the following expressions correctly implements this function?

- [ ] $3(x-2)_+ + 4(x-5)_+$
- [ ] $3(x+2)_+ - 4(x+5)_+$
- [x] $3(x-2)_+ - 4(x-5)_+$
- [ ] $3(x-2)_+ - 1(x-5)_+$

Solution: To represent a piecewise linear function using ReLUs, the weight of each ReLU unit corresponds to the change in slope at that specific kink.

(a) At $x = 2$, the slope changes from 0 to 3. The weight is $3 - 0 = 3$. The term is $3(x-2)_+$.

(b) At $x = 5$, the slope changes from 3 to -1. The weight is $m_{new} - m_{old} = -1 - 3 = -4$. The term is $-4(x-5)_+$.

Therefore, the correct expression is $3(x-2)_+ - 4(x-5)_+$.

Question 28 (GANs) Recall that in a Generative Adversarial Network (GAN), the generator G and discriminator D play a minimax game defined by the function

$$\begin{aligned}\mathcal{L}(D,G) &= \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_z}[\log(1-D(G(z)))] \\ &= \int_x p_{\text{data}}(x)\log D(x)\,dx + \int_z p_z(z)\log(1-D(G(z)))\,dz \\ &= \int_x p_{\text{data}}(x)\log D(x) + p_g(x)\log(1-D(x))\,dx,\end{aligned}$$

where p_{data} is the data distribution, p_z is the noise distribution, and p_g is the distribution induced by the generator G. Let us fix the generator G and consider the optimal discriminator D^* that maximizes $\mathcal{L}(D,G)$ for this fixed G. Assuming $p_{\text{data}}(x)$ and $p_g(x)$ are nonzero for every x, which of the following expressions correctly represents $D^*(x)$?

- [ ] $\frac{p_g(x)}{p_{\text{data}}(x)+p_g(x)}$
- [ ] $\frac{1}{2}$
- [x] $\frac{p_{\text{data}}(x)}{p_{\text{data}}(x)+p_g(x)}$
- [ ] $\frac{p_{\text{data}}(x)}{p_{\text{data}}(x)-p_g(x)}$
- [ ] $\frac{p_{\text{data}}(x)}{p_g(x)}$

Solution: **Correct answer:** $\mathrm{p}_{\text{data}}(x)\frac{}{p_{\text{data}}(x)+p_g(x)Tofindtheoptimaldiscriminator}$D* for a fixed generator G, we need to maximize the function $p_{\text{data}}(x)\log D(x) + p_g(x)\log(1-D(x))$ inside the integral for each x. In other words, if we substitute $D(x)$ with a variable y, we need to maximize the function $f(y) = p_{\text{data}}(x)\log y + p_g(x)\log(1-y)$. Differentiating $f(y)$ once gives us $f'(y) = \frac{p_{\text{data}}(x)}{y} - \frac{p_g(x)}{1-y}$. Differentiating again gives us $f''(y) = -\frac{p_{\text{data}}(x)}{y^2} - \frac{p_g(x)}{(1-y)^2}$, which is always negative for $y \in (0,1)$, indicating that $f(y)$ is concave in this interval. Setting the first derivative to zero, we find the maximum point at $D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x)+p_g(x)}$.

Second part: true/false questions

For each question, mark the box (without erasing) TRUE if the statement is **always true** and the box FALSE if it is **not always true** (i.e., it is sometimes false).

Question 29 (Fairness) In non-degenerate cases, where the sensitive attribute A is correlated with the target variable Y, it is possible to simultaneously satisfy the independence, separation, and sufficiency criteria.

■ FALSE ☐ TRUE

Solution: False. These criteria are mutually exclusive except in degenerate cases.

Question 30 (BERT) BERT is trained using a causal Next Token Prediction objective, similar to GPT models, but unlike GPT models, also optimizes a Next Sentence Prediction objective.

☐ True ■ False

Solution: False. BERT uses Masked Language Modeling (MLM) and Next Sentence Prediction (NSP), which are non-causal and allow the model to see both left and right context. GPT models use causal Next Token Prediction.

Question 31 (Stochastic Gradient Descent) The stochastic gradient used in SGD is a noisy but unbiased estimate of the full gradient.

■ TRUE ☐ FALSE

Solution: True. By sampling a random training example n, we have $\mathbb{E}[\nabla L_n(w)] = \nabla L(w)$; the estimate is unbiased but noisy.

Question 32 (Projection Matrix in Least Squares) In ordinary least squares, the fitted values are given by

$$\hat{y} = X(X^\top X)^{-1}X^\top y = Py,$$

where $P = X(X^\top X)^{-1}X^\top$ is the projection matrix onto the column space of X. The projection matrix P satisfies $P^2 = P$ for any full-rank design matrix X.

■ TRUE ☐ FALSE

Solution: True. The matrix $P = X(X^\top X)^{-1}X^\top$ is always idempotent ($P^2 = P$) because it projects vectors onto the column space of X. This follows from:

$$P^2 = X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top = X(X^\top X)^{-1}X^\top = P.$$

This holds for any full-column-rank X, not just for orthogonal designs.

Question 33 (Bias-Variance Decomposition) When tuning the parameter K in the K-Nearest Neighbors model, if we observe that the model is underfitting (i.e., it is in the 'High Bias' region of the complexity curve), the correct action to increase model complexity is to increase the hyperparameter k.

☐ TRUE ■ FALSE

Solution: The KNN model behaves as followed:

- High k (e.g., $k = N$): A large k means the model averages over many neighbors, which is stable and has low variance but is often too simple and has high bias (underfitting).
- Low k (e.g., $k = 1$): A small k makes the model's prediction highly dependent on just a few (or one) neighboring points. This creates a very flexible, complex model that can fit the training data's noise, leading to low bias but high variance (overfitting).

  Therefore, to increase model complexity and move from the 'High Bias' region to the 'High Variance' region, we must decrease the value of k.

Question 34 (SVM) The cost of training a kernel SVM scales up linearly with the dimensionality of the input features.

■ FALSE ☐ TRUE

Solution: False. The cost of training a kernel SVM scales up with the number of support vectors, not the dimensionality of the input features.

Question 35 (Kernel Methods) Let $\kappa_1(x, x')$ and $\kappa_2(x, x')$ be two valid kernel functions. The function defined by their difference, $\kappa(x, x') = \kappa_1(x, x') - \kappa_2(x, x')$, is guaranteed to be a valid kernel function.

☐ TRUE ■ FALSE

Solution: False. **positive** linear combinations of kernels are valid kernels. Subtracting kernels (negative linear combination) does not guarantee that the resulting Gram matrix is Positive Semi-Definite (PSD), which is a requirement for a valid kernel under Mercer's condition.

Question 36 (Diffusion) In Denoising Score Matching for Diffusion Models, minimizing the error in predicting the added noise ξ_t at time-step t is equivalent to matching the score function $\nabla_x \log p_t(x)$.

☐ False ■ True

Solution: True. Recall that minimizing the DMS loss at time t, $\mathbb{E}_{X_t}[\|s_\theta(t, X_t) - \nabla \log p(X_t)\|_2^2]$, is equivalent to minimizing $\mathbb{E}_{X_0}\mathbb{E}_{X_t|X_0}[\|s_\theta(t, X_t) - \nabla \log p(X_t|X_0)\|_2^2]$. Now, $X_t = \alpha^t X_0 + \sqrt{1-\alpha^{2t}}\xi_t$, where $\xi_t \sim \mathcal{N}(0, I_d)$. Hence, $\nabla \log p(X_t|X_0) = -\frac{\xi_t}{\sqrt{1-\alpha^{2t}}}$, and if we paramtrize $s_\theta(t, X_t)$ as $-\frac{\hat{\xi}_\theta(t,X_t)}{\sqrt{1-\alpha^{2t}}}$, then minimizing the above loss is equivalent to minimizing $\mathbb{E}_{X_0}\mathbb{E}_{X_t|X_0}[\|\hat{\xi}_\theta(t, X_t) - \xi_t\|_2^2]$.

Question 37 (Optimization) The Hessian of the Mean Squared Error (MSE) loss in linear regression is constant with respect to w.

■ TRUE ☐ FALSE

Solution: True. The Hessian is $\frac{1}{N}X^\top X$, which does not depend on w.

Question 38 (Nearest Neighbors) The computational cost of KNN training increases with dataset size.

■ FALSE ☐ TRUE

Solution: False. KNN has no training phase. The computational cost of prediction increases with dataset size.

Question 39 (LLMs) During the inference phase of In-Context Learning, the model's parameters θ are updated to minimize the loss.

■ FALSE ☐ TRUE

Solution: False. In-Context Learning does not modify the model's parameters.

Question 40 (Linear Regression) Adding a new feature to a linear regression model can never increase the training mean squared error.

■ TRUE ☐ FALSE

Solution: True. OLS minimizes training error, so adding a feature can only keep or lower it (never increase)

Question 41 (K-means-PCA) Both the PCA and the k-means problems can be formulated as

$$\min_{W,z_1,\ldots,z_n} \sum_{i=1}^{n} \|Wz_i - x_i\|_2^2,$$

albeit, with different constraints on the matrix W and vectors $z_1, \ldots, z_n$.

■ TRUE ☐ FALSE

Solution: True. Both Principal Component Analysis (PCA) and k-means clustering can be viewed as matrix factorization problems that seek to minimize the reconstruction error (squared Euclidean distance). The objective function for both is:

$$\min_{W,Z} \sum_{i=1}^{n} \|x_i - Wz_i\|_2^2$$

The distinction lies entirely in the constraints applied to the factorization matrices:

(a) **PCA (Dimensionality Reduction):**

- **Matrix W:** The columns represent the principal components (basis vectors). The constraint is that the columns must be **orthonormal** ($W^TW = I$).
- **Vector z_i:** Represents the continuous coordinates (scores) of x_i in the reduced space. There are **no constraints** on the values of z_i (they are essentially W^Tx_i).

(b) **k-Means (Clustering):**

- **Matrix W:** The columns represent the k cluster **centroids** ($\mu_1, \ldots, \mu_k$). There are usually no geometric constraints on their locations.
- **Vector z_i:** Represents the cluster assignment. The constraint is that z_i must be a **one-hot vector** (also called a standard basis vector, e_k). It contains exactly one 1 and zeros elsewhere. This forces the term Wz_i to strictly select one specific column (centroid) from W.

Question 42 (EM) For Gaussian Mixture Models, the estimated cluster centers at convergence of the EM algorithm are independent of the initialization of the cluster centers.

■ FALSE ☐ TRUE

Solution: False. The EM algorithm can converge to different local optima depending on the initialization of the cluster centers.

Third part, open questions

Answer in the empty space below. Your answer should be carefully justified, and all the steps of your argument should be discussed in details. Leave the check-boxes empty, they are used for the grading.

Answer in the space provided! Your answer must be justified with all steps. Leave the check-boxes empty, they are used for the grading.

1 Ridge Regression

Question 43: (*3 points.*) **(Bias of Ridge Regression)**
Consider the ridge regression objective:

$$L_\lambda(w) = \frac{1}{2N}\sum_{n=1}^{N}(y_n - x_n^\top w)^2 + \frac{\lambda}{2}\|w\|_2^2,$$

.

Assume the data are generated according to the linear model

$$y = Xw^* + \varepsilon, \quad \text{with } \varepsilon \sim \mathcal{N}(0, \sigma^2 I_N).$$

Answer the following:

(a) Derive the closed-form expression for the ridge estimator $\hat{w}_\lambda = \arg\min_w L_\lambda(w)$.

(b) Show that the ridge estimator is *biased*, i.e. $\mathbb{E}[\hat{w}_\lambda] \neq w^*$. Derive the expression for $\mathbb{E}[\hat{w}_\lambda]$ and identify the bias term explicitly.

☐0 ☐1 ☐2 ■3

Solution:

(a) (2 point) The ridge solution satisfies $\nabla_w L_\lambda(w) = 0$, giving:

$$\hat{w}_\lambda = (X^\top X + N\lambda I)^{-1}X^\top y.$$

(b) (1 point) Under the model $y = Xw^* + \varepsilon$,

$$\mathbb{E}[\hat{w}_\lambda] = (X^\top X + N\lambda I)^{-1}X^\top X w^*,$$

so

$$\mathbb{E}[\hat{w}_\lambda] - w^* = \big[(X^\top X + N\lambda I)^{-1}X^\top X - I\big]w^*,$$

which is nonzero for $\lambda > 0$, showing that the ridge estimator is biased.

2 Optimization with Momentum

Question 44: (*3 points.*) **(Variance Reduction by Momentum)**
Consider the stochastic gradient descent (SGD) with momentum updates:

$$m_{t+1} = \beta m_t + (1-\beta)g_t, \qquad w_{t+1} = w_t - \gamma m_{t+1},$$

where $0 \leq \beta < 1$, $\gamma > 0$, and g_t denotes the stochastic gradient at step t.
Assume that the random gradients $\{g_t\}$ are i.i.d. samples of a random variable g satisfying

$$\mathbb{E}[g] = \mu, \quad \mathrm{Var}[g] = \sigma^2.$$

Answer the following:

(a) Derive the unrolled expression of m_{t+1} as a function of all past gradients $\{g_t, g_{t-1}, \dots, g_0\}$.
(b) Using the independence of the gradients, express $\mathrm{Var}[m_{t+1}]$ in terms of $\mathrm{Var}[g]$, β, and t.
(c) Show that, in the steady-state limit $t \to \infty$,

$$\boxed{\mathrm{Var}[m_t] = \frac{1-\beta}{1+\beta}\,\mathrm{Var}[g].}$$

Briefly interpret this result: how does momentum affect the variance of the gradient estimates?

☐0 ☐1 ☐2 ■3

Solution:

(a) (1 point) Unrolling the recursion:

$$m_{t+1} = (1-\beta)\sum_{k=0}^{t} \beta^k g_{t-k}.$$

Thus, m_{t+1} is an exponentially weighted moving average of all past stochastic gradients.

(b) (1 point) Since the g_t are independent and identically distributed, variances add:

$$\mathrm{Var}[m_{t+1}] = (1-\beta)^2 \sum_{k=0}^{t} \beta^{2k}\,\mathrm{Var}[g] = \sigma^2(1-\beta)^2 \frac{1-\beta^{2(t+1)}}{1-\beta^2}.$$

(c) (1 point) At steady state ($t \to \infty$):

$$\mathrm{Var}[m_t] = \sigma^2 \frac{(1-\beta)^2}{1-\beta^2} = \sigma^2 \frac{1-\beta}{1+\beta}.$$

Therefore, the variance of the momentum term is smaller than the variance of individual gradients by the factor $\frac{1-\beta}{1+\beta} < 1$. Momentum thus acts as an exponential smoother, reducing gradient noise and stabilizing the learning trajectory.

3 Maximum Likelihood Estimator

Question 45: (*3 points.*) **(MLE)**

Let X be a random variable following a uniform distribution on the interval $[0, \theta]$, denoted as $X \sim U(0, \theta)$, where $\theta > 0$ is an unknown parameter. The probability density function (PDF) is given by:

$$f(x;\theta) = \begin{cases} \frac{1}{\theta} & \text{if } 0 \le x \le \theta \\ 0 & \text{otherwise} \end{cases}$$

You observe a dataset $\mathcal{D} = \{1, 3, 6, 3, 4\}$ consisting of $n = 5$ independent and identically distributed (i.i.d.) samples.

(a) Write down the likelihood function $\mathcal{L}(\theta; \mathcal{D})$ for the parameter θ given the observed data. Be precise regarding the constraints on θ.
(b) Derive the Maximum Likelihood Estimator (MLE) $\hat{\theta}_{MLE}$ for general data points $x_1, \dots, x_n$. Explain why the standard method of setting the derivative of the log-likelihood to zero does not apply here.
(c) Calculate the specific numerical value of $\hat{\theta}_{MLE}$ for the dataset $\mathcal{D}$.

☐0 ☐1 ☐2 ■3

Solution:

(a) (1 point) The likelihood function is the joint probability density of the observed data viewed as a function of the parameter θ. Since the samples are i.i.d., we multiply the individual PDFs:

$$\mathcal{L}(\theta;\mathcal{D}) = \prod_{i=1}^{n} f(x_i;\theta)$$

Substituting the PDF of the uniform distribution:

$$\mathcal{L}(\theta;\mathcal{D}) = \prod_{i=1}^{n} \left(\frac{1}{\theta} \cdot \mathbb{I}(0 \le x_i \le \theta)\right)$$

where $\mathbb{I}(\cdot)$ is the indicator function. This can be simplified as:

$$\mathcal{L}(\theta;\mathcal{D}) = \begin{cases} \frac{1}{\theta^n} & \text{if } \theta \ge \max(x_1,\dots,x_n) \text{ and } \min(x_1,\dots,x_n) \ge 0 \\ 0 & \text{otherwise} \end{cases}$$

Essentially, if θ is smaller than any of our observations (e.g., if we observe a 6 but guess $\theta = 5$), the probability of that observation is 0, making the likelihood 0.

(b) (1 point) We look for the value of θ that maximizes $\mathcal{L}(\theta) = \frac{1}{\theta^n}$.

If we attempt to take the log-likelihood $\ell(\theta) = -n\ln(\theta)$ and differentiate with respect to θ, we get $\frac{d\ell}{d\theta} = -\frac{n}{\theta}$. Setting this to 0 yields no solution (or $\theta \to \infty$, which minimizes the likelihood rather than maximizing it). This indicates that the maximum lies on the boundary of the domain, not at a critical point.

Since $\frac{1}{\theta^n}$ is a strictly decreasing function for $\theta > 0$, we want to choose the smallest possible value for θ that remains valid. From step 1, we know the constraint is $\theta \ge \max(x_1,\dots,x_n)$.

Therefore, the smallest valid θ (which maximizes the likelihood) is the sample maximum:

$$\hat{\theta}_{MLE} = \max(x_1, x_2, \dots, x_n)$$

(c) (1 point) For the dataset $\mathcal{D} = \{1, 3, 6, 3, 4\}$, we find the maximum value:

$$\hat{\theta}_{MLE} = \max(1, 3, 6, 3, 4) = 6$$

4 Expectation Maximization Algorithm

Question 46: (*8 points.*) **(EM for Light Bulbs)**

We use the (soft) Expectation Maximization (EM) algorithm to compute a Maximum Likelihood Estimator (MLE) for the average lifetime of light bulbs. PDF is given by: $p(x|\theta) = \frac{1}{\theta}e^{-\frac{x}{\theta}}$).

We test $N + M$ independent light bulbs in two independent experiments. We test the first N light bulbs and observe their exact lifetimes. Let $Y = (Y_1, \dots, Y_N)$ denote these lifetimes. We test the remaining M bulbs, but we only check them at a fixed time $t > 0$. Let $X = (X_1, \dots, X_M)$ be the observations, where $X_j = 1$ if the bulb was still working at time t (censored), and $X_j = 0$ if it had already expired. Let $Z = (Z_1, \dots, Z_M)$ denote the *unobserved* exact lifetimes of these bulbs.

(a) (2 points) Write down the complete-data log-likelihood $\log p(X, Y, Z|\theta)$.

(b) (2 points) In the E-step, we compute the expected lifetime of a bulb that is still working at time t. Calculate $E_1(\theta') \triangleq \mathbb{E}[Z_j|X_j = 1, \theta']$.

(c) (2 points) In the E-step, we also compute the expected lifetime of a bulb that has already expired by time t. Calculate $E_0(\theta') \triangleq \mathbb{E}[Z_j|X_j = 0, \theta']$.

(d) (2 points) Define $k = \sum_{j=1}^{M} \mathbb{I}_{\{X_j=1\}}$ as the number of working bulbs in the second experiment. Write down the expected complete data log-likelihood function $Q(\theta, \theta') \triangleq \mathbb{E}_Z[\log p(X, Y, Z|\theta)|X, Y, \theta']$. express your answer using $N, M, k, E_1(\theta')$, and $E_0(\theta')$.

0 1 2 3

Solution:

(a) (2 points) The likelihood of independent exponential variables is the product of their PDFs. The PDF is $f(x) = \frac{1}{\theta}e^{-x/\theta}$.

The joint PDF for Y and Z is:

$$p(Y, Z|\theta) = \prod_{i=1}^{N} \frac{1}{\theta}e^{-Y_i/\theta} \prod_{j=1}^{M} \frac{1}{\theta}e^{-Z_j/\theta}$$

(Note: X is deterministic given Z, so it does not add a term to the complete likelihood density). Taking the log:

$$\log p(X, Y, Z|\theta) = -(N + M)\log\theta - \frac{1}{\theta}\sum_{i=1}^{N} Y_i - \frac{1}{\theta}\sum_{j=1}^{M} Z_j$$

(b) (2 points) We need $\mathbb{E}[Z_j|Z_j > t]$. Using the *memoryless property* of the exponential distribution, if a component has survived until time t, its remaining life is distributed as the original exponential. Thus, the total expected life is the time already passed (t) plus the original mean (θ'). Alternatively, by integration: $\int_t^\infty z\frac{1}{\theta'}e^{-(z-t)/\theta'}dz = t + \theta'$.

$$E_1(\theta') = \theta' + t$$

(c) (2 points) We need $\mathbb{E}[Z_j|Z_j \le t]$. Using the definition of conditional expectation:

$$\mathbb{E}[Z|Z \le t] = \frac{\int_0^t x\frac{1}{\theta'}e^{-x/\theta'}dx}{P(Z \le t)}$$

The denominator is $1 - e^{-t/\theta'}$. The numerator is $\theta'(1 - e^{-t/\theta'}) - te^{-t/\theta'}$ (via integration by parts). Dividing gives:

$$E_0(\theta') = \theta' - \frac{te^{-t/\theta'}}{1 - e^{-t/\theta'}}$$

(d) (2 points) The Q function is the expectation of the log-likelihood (from Step 1) with respect to the missing data Z, conditioned on the current parameters θ' and observed data X.

Since the log-likelihood is linear in Z_j, we replace each Z_j with its conditional expectation. For the k bulbs where $X_j = 1$, the expectation is $E_1(\theta')$. For the $M - k$ bulbs where $X_j = 0$, the expectation is $E_0(\theta')$.

$$Q(\theta, \theta') = -(N + M)\log\theta - \frac{1}{\theta}\sum_{i=1}^{N} Y_i - \frac{1}{\theta}(kE_1(\theta') + (M - k)E_0(\theta'))$$

5 Backpropagation

Question 47: (*7 points.*) Let $x = (x_1, x_2, \ldots, x_d)^\top \in \mathbb{R}^d$ denote a *non-constant* d-dimensional input vector. We consider a neural network with *Layer Normalization* (LayerNorm) as the first layer, followed by a differentiable function $h : \mathbb{R}^d \to \mathbb{R}^k$, and finally a scalar loss function $\ell : \mathbb{R}^k \to \mathbb{R}$. The overall mapping is

$$\mathcal{L}(x) = \ell\big(h(\mathrm{LN}(x))\big) \in \mathbb{R},$$

where LN represents the LayerNorm transformation with learnable parameters $\gamma, \beta \in \mathbb{R}^d$, computed as

$$\mathrm{LN}(x)_i = \gamma_i \frac{x_i - \mu(x)}{\sigma(x)} + \beta_i, \quad i \in \{1, \ldots, d\}.$$

where

$$\mu(x) = \frac{1}{d}\sum_{i=1}^{d} x_i, \quad v(x) = \frac{1}{d}\sum_{i=1}^{d}(x_i - \mu(x))^2, \quad \sigma(x) = \sqrt{v(x)}$$

denote the *empirical* mean, variance and standard deviation of x.

- **Forward pass (1 point):** Let $z = ax + b$, where $a > 0, a \in \mathbb{R}, b \in \mathbb{R}$. Show that $\mathrm{LN}(x) = \mathrm{LN}(z)$.
- **Backward pass (2 points):** Find the analytical expressions for $\frac{\partial \mathcal{L}(x)}{\partial \gamma}$ and $\frac{\partial \mathcal{L}(x)}{\partial \beta}$, assuming you are given $\delta_i = \frac{\partial \mathcal{L}(x)}{\partial y_i}$, where $y = \mathrm{LN}(x)$. Your final answer must be written in terms of δ, x, γ, β, where $\delta = (\delta_1, \ldots, \delta_d)^\top$.
- **Gradient after scaling (2 points):** Still assuming that $z = ax + b$ with $a > 0$. Write down the gradients $\frac{\partial \mathcal{L}(z)}{\partial \beta}$ and $\frac{\partial \mathcal{L}(z)}{\partial \gamma}$ when z is the input, in terms of $\frac{\partial \mathcal{L}(x)}{\partial \beta}$ and $\frac{\partial \mathcal{L}(x)}{\partial \gamma}$, the gradients when x is the input. How do these gradient relate to each other?
- **Relationship between gradients (1 point):** Let $g_x = \frac{\partial \mathcal{L}(x)}{\partial x}$ denote the gradient of the loss with respect to the input x, and let $g_z = \frac{\partial \mathcal{L}(z)}{\partial z}$ denote the gradient with respect to z. Using previous results, check whether these two gradients are equal, and if not, whether they can be expressed in terms of each other.
- **Orthogonality (1 point):** We found that $\mathcal{L}(ax) = \mathcal{L}(x)$, for $a > 0$. Let $g_w = \frac{\partial \mathcal{L}(w)}{\partial w}$ denote the gradient of the loss with respect to w, an arbitrary input. Show that g_w is orthogonal to w. Equivalently, show that the dot product $w^\top g_w = 0$. Hint: define $g(a) = \mathcal{L}(aw)$ and differentiate with respect to a. How does g look on a plot? Note: this property holds for *any* differentiable function such that $\mathcal{L}(ax) = \mathcal{L}(x)$, not only in this exam question.

☐ 0 ☐ 1 ☐ 2 ■ 3

Solution:

(a) Since $z_i = ax_i + b$, we have

$$\mu(z) = \frac{1}{d}\sum_{i=1}^{d} z_i = \frac{1}{d}\sum_{i=1}^{d}(ax_i + b) = a\mu(x) + b, \tag{1}$$

$$v(z) = \frac{1}{d}\sum_{i=1}^{d}(z_i - \mu(z))^2 = \frac{1}{d}\sum_{i=1}^{d} a^2(x_i - \mu(x))^2 = a^2 v(x). \tag{2}$$

Since $a > 0$,

$$\sigma(z) = \sqrt{v(z)} = \sqrt{a^2 v(x)} = a\sigma(x). \tag{3}$$

Thus

$$\mathrm{LN}(z)_i = \gamma_i \frac{z_i - \mu(z)}{\sigma(z)} + \beta_i \tag{4}$$

$$= \gamma_i \frac{(ax_i + b) - (a\mu(x) + b)}{a\sigma(x)} + \beta_i \tag{5}$$

$$= \gamma_i \frac{x_i - \mu(x)}{\sigma(x)} + \beta_i = \mathrm{LN}(x)_i, \tag{6}$$

so $\mathrm{LN}(x) = \mathrm{LN}(z)$.

(b) From the layer normalization equation with input x,

$$y_i = \gamma_i \frac{x_i - \mu(x)}{\sigma(x)} + \beta_i \tag{7}$$

and the chain rule, since $\frac{\partial y_i}{\partial \beta_i} = 1$ we get

$$\frac{\partial \mathcal{L}(x)}{\partial \beta_i} = \frac{\partial \mathcal{L}(x)}{\partial y_i} \cdot \frac{\partial y_i}{\partial \beta_i} = \delta_i \tag{8}$$

and since $\frac{\partial y_i}{\partial \gamma_i} = \frac{x_i - \mu(x)}{\sigma(x)}$ we obtain

$$\frac{\partial \mathcal{L}(x)}{\partial \gamma_i} = \frac{\partial \mathcal{L}(x)}{\partial y_i} \cdot \frac{\partial y_i}{\partial \gamma_i} = \delta_i \cdot \frac{x_i - \mu(x)}{\sigma(x)} \tag{9}$$

In vector form,

$$\frac{\partial \mathcal{L}(x)}{\partial \beta} = \delta, \quad \frac{\partial \mathcal{L}(x)}{\partial \gamma} = \delta \odot \frac{x - \mu(x)}{\sigma(x)} \tag{10}$$

where $\odot$ denotes element-wise multiplication.

(c) Let $y = \mathrm{LN}(x)$ and $y' = \mathrm{LN}(z)$, and define

$$\delta_i = \frac{\partial \mathcal{L}(x)}{\partial y_i}, \qquad \delta_i' = \frac{\partial \mathcal{L}(z)}{\partial y_i'}. \tag{11}$$

From Question 1 we know that for $z = ax + b$ with $a > 0$,

$$\mathrm{LN}(z) = \mathrm{LN}(x), \tag{12}$$

so $y_i' = y_i$ for all i. Therefore, by the chain rule,

$$\delta_i' = \frac{\partial \mathcal{L}(z)}{\partial y_i'} = \frac{\partial \mathcal{L}(x)}{\partial y_i} = \delta_i. \tag{13}$$

From Solution 2, when the input is x we have

$$\frac{\partial \mathcal{L}(x)}{\partial \beta_i} = \delta_i, \qquad \frac{\partial \mathcal{L}(x)}{\partial \gamma_i} = \delta_i \cdot \frac{x_i - \mu(x)}{\sigma(x)}. \tag{14}$$

When the input is z, the same derivation gives

$$\frac{\partial \mathcal{L}(z)}{\partial \beta_i} = \delta_i', \qquad \frac{\partial \mathcal{L}(z)}{\partial \gamma_i} = \delta_i' \cdot \frac{z_i - \mu(z)}{\sigma(z)}. \tag{15}$$

Using $\delta_i' = \delta_i$, $\mu(z) = a\mu(x) + b$ and $\sigma(z) = a\sigma(x)$, we compute

$$\begin{aligned}
\frac{z_i - \mu(z)}{\sigma(z)} &= \frac{(ax_i + b) - (a\mu(x) + b)}{a\sigma(x)} && (16)\\
&= \frac{a(x_i - \mu(x))}{a\sigma(x)} && (17)\\
&= \frac{x_i - \mu(x)}{\sigma(x)}. && (18)
\end{aligned}$$

Therefore,

$$\frac{\partial \mathcal{L}(z)}{\partial \beta_i} = \delta_i = \frac{\partial \mathcal{L}(x)}{\partial \beta_i}, \qquad \frac{\partial \mathcal{L}(z)}{\partial \gamma_i} = \delta_i \cdot \frac{x_i - \mu(x)}{\sigma(x)} = \frac{\partial \mathcal{L}(x)}{\partial \gamma_i}. \tag{19}$$

so the gradients with respect to β and γ are *invariant* to scaling and shifting of the input.

(d) Recall that $z = ax + b$ Therefore, by the chain rule,

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial \mathcal{L}}{\partial z_i}\frac{\partial z_i}{\partial x_i} = \frac{\partial \mathcal{L}}{\partial z_i} a. \tag{20}$$

Since $a > 0$, we can rewrite

$$\frac{\partial \mathcal{L}}{\partial z_i} = \frac{1}{a}\frac{\partial \mathcal{L}}{\partial x_i}. \tag{21}$$

Hence, the backward pass is *not* invariant to the input.

(e) Define the scalar-valued function $g(a) = \mathcal{L}(aw)$ for $a > 0$. By the chain rule,

$$g'(a) = \nabla_w \mathcal{L}(aw)^\top w, \tag{22}$$

where $\nabla_w \mathcal{L}(aw)$ denotes the gradient of $\mathcal{L}$ with respect to its argument, evaluated at aw.

Since $\mathcal{L}(aw) = \mathcal{L}(w)$ for all $a > 0$, $g(a)$ is constant, so $g'(a) = 0$ for all a. Evaluating at $a = 1$ gives

$$0 = g'(1) = \nabla_w \mathcal{L}(w)^\top w = w^\top g_w, \tag{23}$$

where in the last equality we used the definition $g_w = \nabla_w \mathcal{L}(w)$. Hence w is orthogonal to g_w.

6 Post-training & Alignment

Question 48: (*3 points.*) In the Proximal Policy Optimization (PPO) algorithm used for Reinforcement Learning from Human Feedback (RLHF), we typically optimize the following objective function:

$$\arg\max_\theta \mathbb{E}_{x_{\text{prompt}} \sim \mathcal{D}, x \sim \hat{\pi}_\theta} \left[\frac{\pi_\theta(x|x_{\text{prompt}})}{\hat{\pi}_\theta(x|x_{\text{prompt}})} R_\phi(x_{\text{prompt}}, x) - \beta \cdot D_{KL}(\hat{\pi}_\theta(x|x_{\text{prompt}}) || \pi_\theta(x|x_{\text{prompt}})) \right]$$

where $\hat{\pi}_\theta$ denotes the old policy used to collect the data, and π_θ is the current policy being optimized.

(a) Explain the role of the ratio $\frac{\pi_\theta(x|x_{\text{prompt}})}{\hat{\pi}_\theta(x|x_{\text{prompt}})}$ and why it's necessary.

(b) Explain what the KL-Divergence measures and why it's necessary.

(c) Describe what happens to the optimization behavior when β is too small versus too large, and explain why finding the right balance is crucial for successful RLHF.

☐ 0 ☐ 1 ☐ 2 ■ 3

Solution:

(a) **Importance Sampling Ratio (1 points):** The ratio $\frac{\pi_\theta(x|x_{\text{prompt}})}{\hat{\pi}_\theta(x|x_{\text{prompt}})}$ corrects for the distribution mismatch between sampling and optimization. In PPO, we perform multiple optimization steps using data collected by an "old" version of the policy ($\hat{\pi}_\theta$) to improve sample efficiency. Since the expectation is computed over samples from $\hat{\pi}_\theta$, but we want to optimize π_θ, this ratio acts as an importance sampling weight. It gives higher weight to actions that the current policy π_θ prefers relative to the old policy, and lower weight to actions it would avoid.

(b) **Constraint Role (1 point):** The KL term acts as a Trust Region constraint. It penalizes the new policy π_θ from diverging too far from the behavior policy $\hat{\pi}_\theta$ that collected the data. This ensures the importance sampling approximation remains valid and the optimization is stable.

(c) **Risk of Large Updates (1 points): Too small β (Weak Constraint):** If the constraint is too weak, the optimizer may take excessively large steps. This leads to destructive updates, causing "catastrophic forgetting" of the pre-trained knowledge.

Too large β (Strong Constraint): The model cannot learn effectively because it is forced to stay nearly identical to the old policy.

7 Bias-Variance Decomposition

Question 49: (*2 points.*) **(Bias Variance Decomposition)**

Consider a regression problem where the true underlying function is given by:

$$f(x) = 3x^2$$

We have a learning algorithm that generates a prediction model based on a **single** training example. The training process is defined as follows:

- A single training point, x_{train}, is sampled from a Uniform distribution over the interval $[0, 1]$.
- The label is observed without noise, such that $y_{train} = f(x_{train})$.
- The learning algorithm produces a model $\hat{f}(x)$ that acts as a **constant predictor**, meaning it predicts the value of the observed label for all future inputs. Formally:

$$\hat{f}(x) = y_{train} = 3(x_{train})^2$$

Based on the definition of the prediction model, answer the following questions for the specific query point $x_0 = 0.5$:

(a) (1 point) Calculate the **Bias** of the prediction model.

(b) (1 point) Calculate the **Variance** of the prediction model.

☐0 ☐1 ☐2 ■3

Solution:

(a) (1 point) First, we calculate the true value of the function at the query point $x_0 = 0.5$:

$$f(x_0) = 3(0.5)^2 = 3(0.25) = 0.75$$

Then, we calculate the expected prediction $\mathbb{E}[\hat{f}(x_0)]$.

$$\mathbb{E}[\hat{f}(x_0)] = \mathbb{E}[3X^2] = 3\int_0^1 x^2\,dx = 3\left[\frac{x^3}{3}\right]_0^1 = 3\left(\frac{1}{3}\right) = 1$$

The Bias is defined as the difference between the expected prediction and the true value:

$$\text{Bias}(\hat{f}(x_0)) = \mathbb{E}[\hat{f}(x_0)] - f(x_0) = 1 - 0.75 = \mathbf{0.25}$$

(b) (1 point) The Variance is defined as $\mathbb{E}[(\hat{f}(x_0) - \mathbb{E}[\hat{f}(x_0)])^2]$. We can use the computational formula:

$$\text{Var}[\hat{f}(x_0)] = \mathbb{E}[\hat{f}(x_0)^2] - (\mathbb{E}[\hat{f}(x_0)])^2$$

From the calculations of the previous section we know that $(\mathbb{E}[\hat{f}(x_0)])^2 = 1$. We just need to find the second moment $\mathbb{E}[\hat{f}(x_0)^2]$:

$$\mathbb{E}[\hat{f}(x_0)^2] = \int_0^1 (3x^2)^2\,dx = \int_0^1 9x^4\,dx = 9\left[\frac{x^5}{5}\right]_0^1 = 9\left(\frac{1}{5}\right) = 1.8$$

Putting it all together, we achieve $\text{Var}[\hat{f}(x_0)] = 1.8 - 1 = 0.8$

8 ReLU Network and Symmetric Initialization

Question 50: (*5 points.*) **(Affine form at symmetric initialization)**

Let $\sigma : \mathbb{R}^n \to \mathbb{R}^n$ be the componentwise ReLU,

$$\sigma(x)_i = \max(0, x_i).$$

Let

$$f(z) = W_f z + b_f, \qquad g(z) = W_g z + b_g$$

be affine maps with $W_f, W_g \in \mathbb{R}^{m\times n}$ and $b_f, b_g \in \mathbb{R}^m$. Define

$$h(x) = f\big(\sigma(x)\big) + g\big(-\sigma(-x)\big), \qquad x \in \mathbb{R}^n.$$

Assume symmetric initialization $W_f = W_g =: W$ and $b_f = b_g =: b$.

Task. *Prove from first principles that h is an affine function of x; i.e. show that there exist $W_{\text{eff}} \in \mathbb{R}^{m\times n}$ and $b_{\text{eff}} \in \mathbb{R}^m$ such that for all $x \in \mathbb{R}^n$,*

$$h(x) = W_{\text{eff}}x + b_{\text{eff}}.$$

Prove any scalar identity you need from first principles. (5 points)

□0 □1 □2 ■3

Solution: Step 1: Algebraic manipulation of h.
Let $x^+ := \sigma(x)$ and $x^- := \sigma(-x)$. Using $W_f = W_g = W$ and $b_f = b_g = b$,

$$\begin{aligned} h(x) &= f(\sigma(x)) + g(-\sigma(-x)) \\ &= (Wx^+ + b) + (W(-x^-) + b) \\ &= W(x^+ - x^-) + 2b. \end{aligned}$$

Thus it suffices to prove the componentwise identity $x = x^+ - x^-$.

Step 2: Scalar identity $t = t^+ - t^-$.
For a scalar $t \in \mathbb{R}$ define

$$t^+ := \max(0, t), \qquad t^- := \max(0, -t).$$

We consider two exhaustive cases:

- If $t \geq 0$, then $t^+ = t$ and $t^- = 0$, hence $t^+ - t^- = t$.
- If $t < 0$, then $t^+ = 0$ and $t^- = -t$, hence $t^+ - t^- = 0 - (-t) = t$.

Therefore $t = t^+ - t^-$ for every $t \in \mathbb{R}$. Applying this identity coordinatewise to x yields $x = x^+ - x^-$.

Step 3: Conclusion.
Substitute $x^+ - x^- = x$ into the expression for h:

$$h(x) = W(x^+ - x^-) + 2b = Wx + 2b.$$

Hence h is affine with

$$W_{\text{eff}} = W, \qquad b_{\text{eff}} = 2b.$$

□

9 Diffusion Models

Question 51: (*4 points.*) **(Forward Process Marginals)** Consider a Diffusion Probabilistic Model. The forward process is defined as a Markov chain where noise is gradually added to the data $X_0 \sim p_0$.The transition kernel at step t is given by:

$$p(X_t|X_{t-1}) = \mathcal{N}(X_t; \sqrt{\alpha_t}X_{t-1}, (1-\alpha_t)I),$$

where $\alpha_t \in (0,1)$ and I is the $d-$dimensional identity matrix. Let $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. By reparametrizing the random variables X_t as sums of independent scaled standard Gaussians, derive a closed-form expression for the marginal distribution $p(X_t|X_0)$.

□0 □1 □2 □3 ■4

Solution: (1 point) Single-step reparameterization: We can re-write X_t using the reparameterization trick as $X_t = \sqrt{\alpha_t}X_{t-1} + \sqrt{1-\alpha_t}\xi_{t-1}$, where $\xi_{t-1} \sim \mathcal{N}(0, I)$.
(2 points) Recursive substitution and merging Gaussians: Now we substitute X_{t-1} using the same logic for the previous step ($X_{t-1} = \sqrt{\alpha_{t-1}}X_{t-2} + \sqrt{1-\alpha_{t-1}}\xi_{t-2}$):

$$\begin{aligned} X_t &= \sqrt{\alpha_t}(\sqrt{\alpha_{t-1}}X_{t-2} + \sqrt{1-\alpha_{t-1}}\xi_{t-2}) + \sqrt{1-\alpha_t}\xi_{t-1} \\ &= \sqrt{\alpha_t\alpha_{t-1}}X_{t-2} + \sqrt{\alpha_t(1-\alpha_{t-1})}\xi_{t-2} + \sqrt{1-\alpha_t}\xi_{t-1}. \end{aligned}$$

Since ξ_{t-2} and ξ_{t-1} are independent Gaussian variables, their weighted sum is also Gaussian. We merge the noise terms using the property $\mathcal{N}(0, \sigma_1^2 I) + \mathcal{N}(0, \sigma_2^2 I) = \mathcal{N}(0, (\sigma_1^2 + \sigma_2^2)I)$. Hence, the combined variance is $1 - \alpha_t\alpha_{t-1}$. Thus, $X_t = \sqrt{\alpha_t\alpha_{t-1}}X_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\bar{\xi}$, where $\bar{\xi} \sim \mathcal{N}(0, I)$.

(1 point) Induction: By induction down to X_0, the coefficient for the mean accumulates the product of alphas ($\sqrt{\bar{\alpha}_t}$), and the variance term becomes $1 - \bar{\alpha}_t$.

$$X_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{1 - \bar{\alpha}_t} \xi,$$

where ξ is a standard Gaussian. This corresponds to the distribution:

$$q(X_t | X_0) = \mathcal{N}(X_t; \sqrt{\bar{\alpha}_t} X_0, (1 - \bar{\alpha}_t) I)$$